

Forecasting Public Interest in "Flu Symptoms" Searches — A Time Series Analysis

Khin Yadanar Aung
66132500216

Advanced Topics in Data Analytics

- Problem Definition and Data Collection
- Exploratory Data Analysis & Visualization
- Preprocessing & Feature Engineering
- Data Splitting
- Model Selection & Training
- Model Evaluation
- Forecasting & Deployment
- Conclusion

Problem Definition

Objective

- To forecast weekly "flu symptoms" search interest
- Explore Google Trends as an early indicator of flu activity

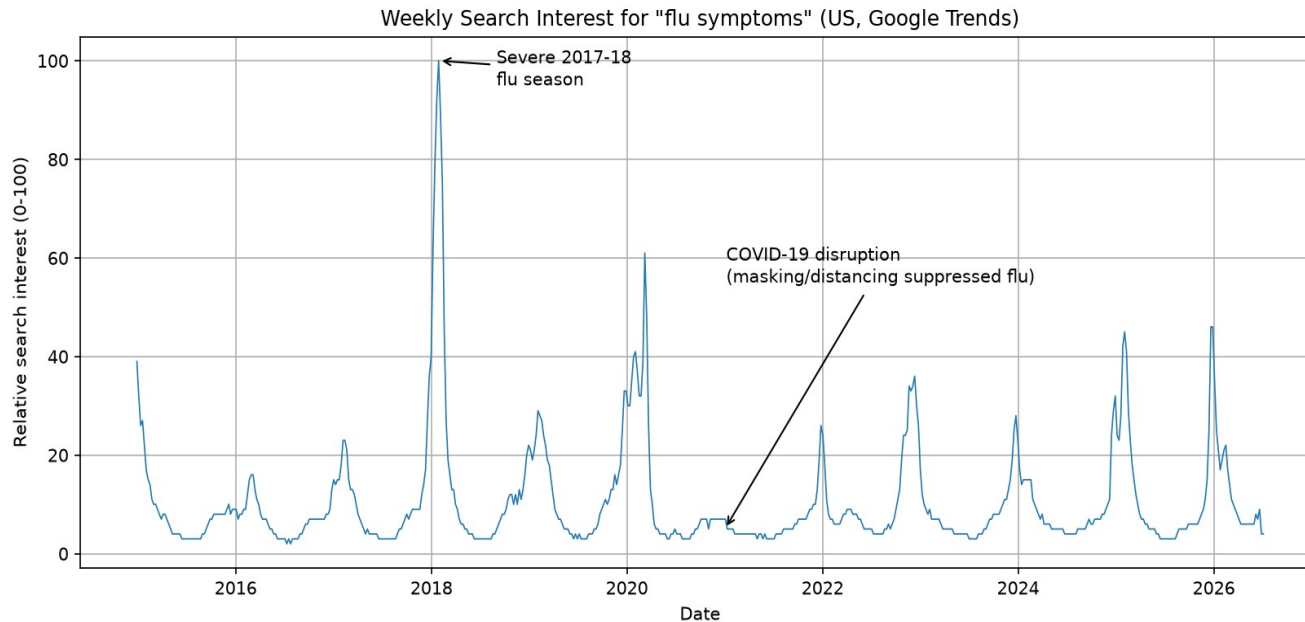
Dataset

- Google Trends (PyTrends API) , at United States
- Dec 2014 – Jul 2026, a total of 5years with 602 weekly observations

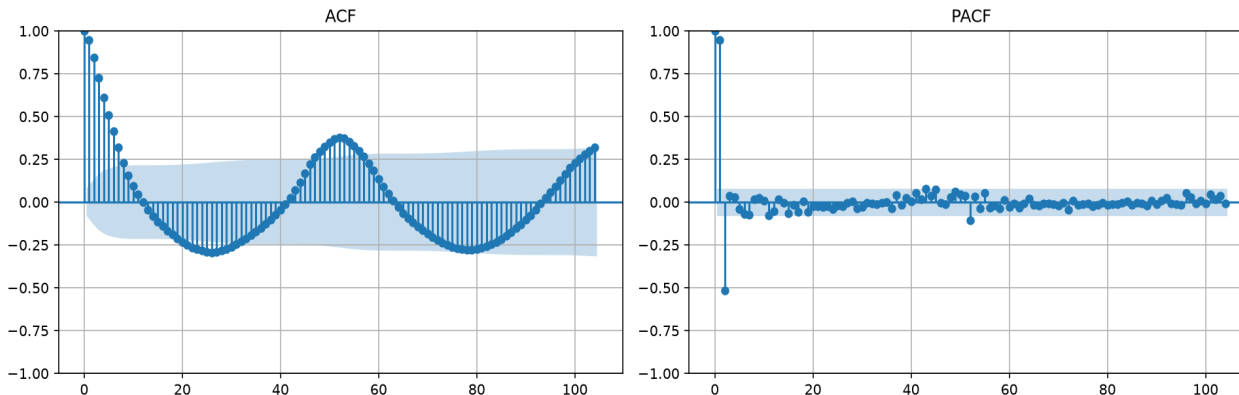
Data Preparation

- Combined 3 overlapping datasets
- Rescaled using overlapping periods
- Created one continuous weekly time series

EDA & Visualization



- Strong yearly seasonal pattern – winter peaks each year
- Highest peak : during the 2017–2018 flu season
- COVID-19 disrupted normal flu activity
- ACF/PACF: slow decay + spike near lag 52 with non-stationary pattern
- ADF tests stationary

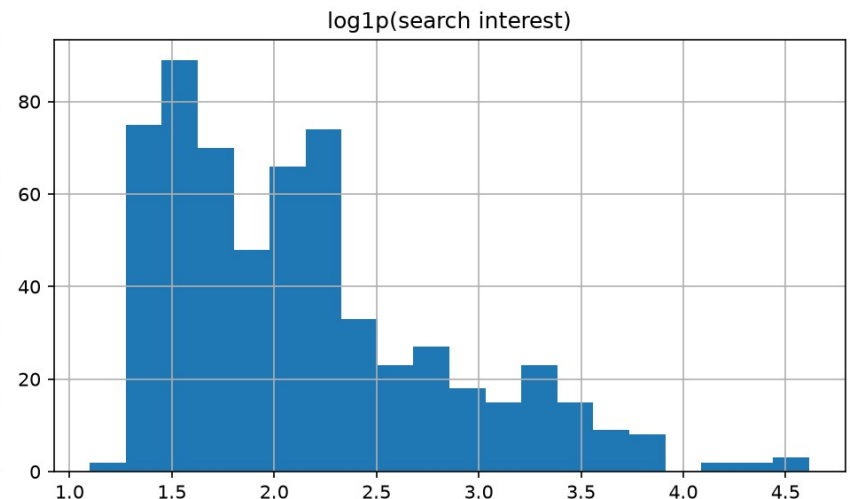
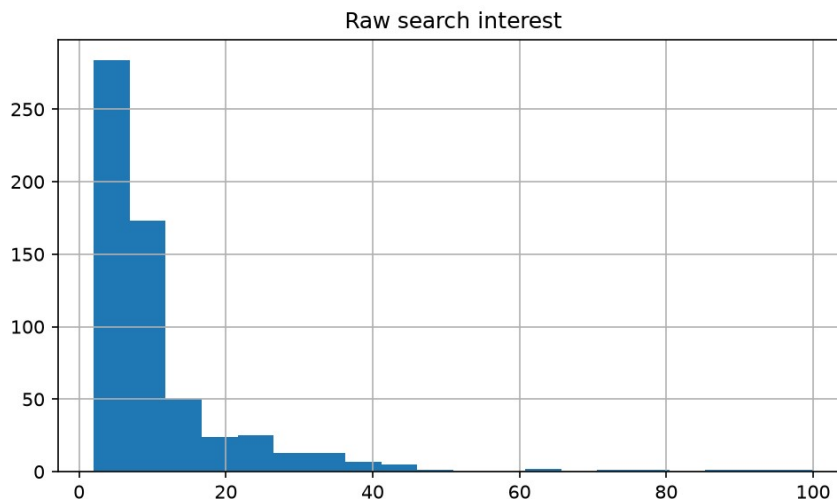


Preprocessing

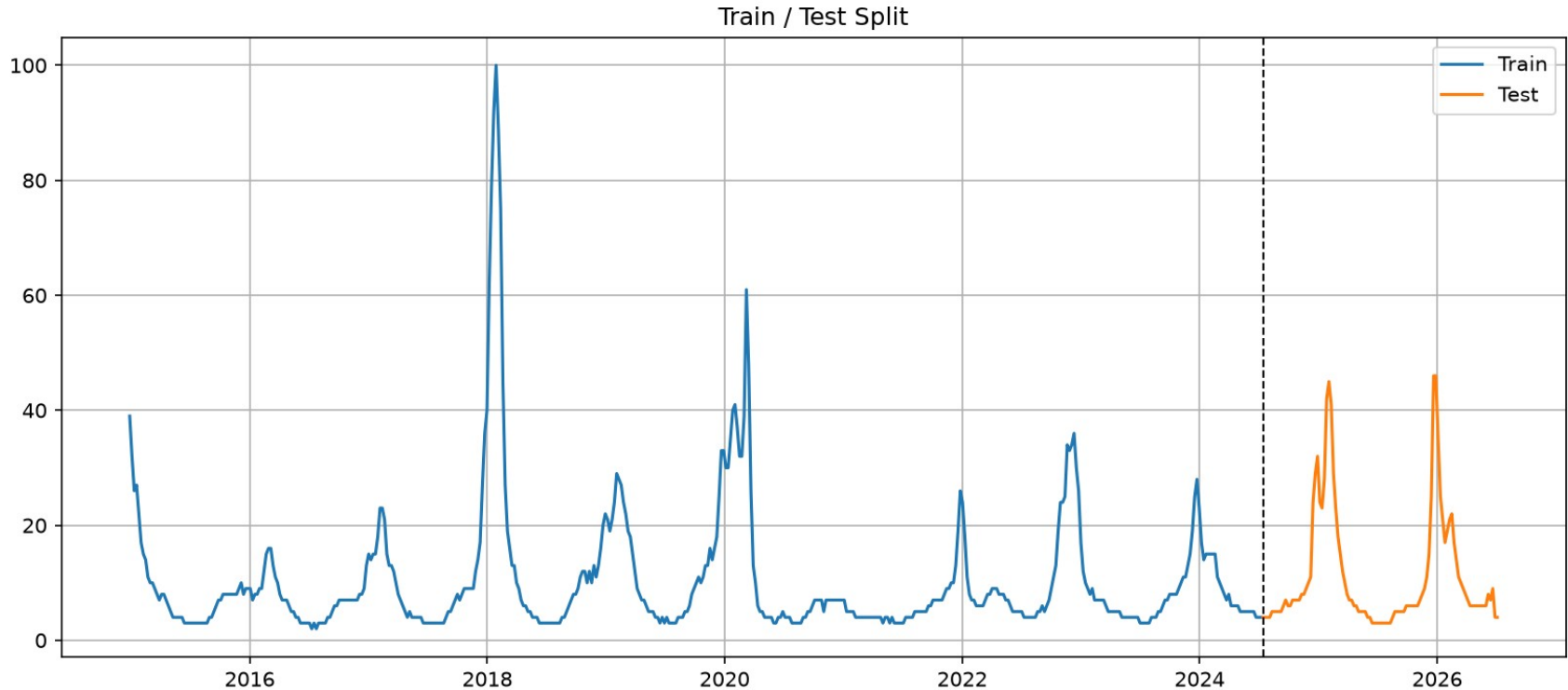
- No missing values
- Applied log1p transformation
- Regular & seasonal differencing
- Stationary verified using ADF

Feature Engineering

- Cyclical features
 - day-of-year (sin)
 - day-of-year (cos)

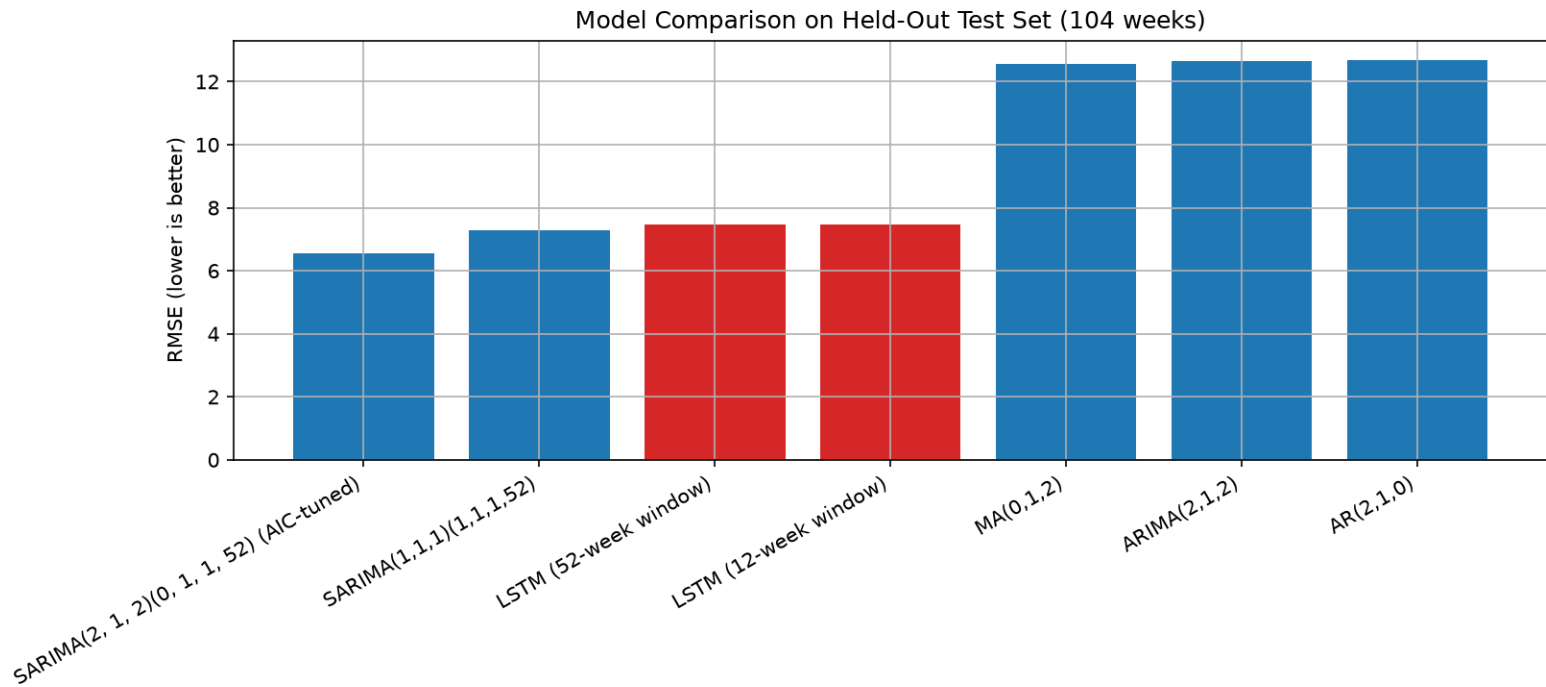


Data Splitting



- Chronological split, no shuffling
- Train: 498 weeks (Dec 2014 – Jul 2024)
- Test: last 104 weeks / ~2 years (Jul 2024 – Jul 2026)

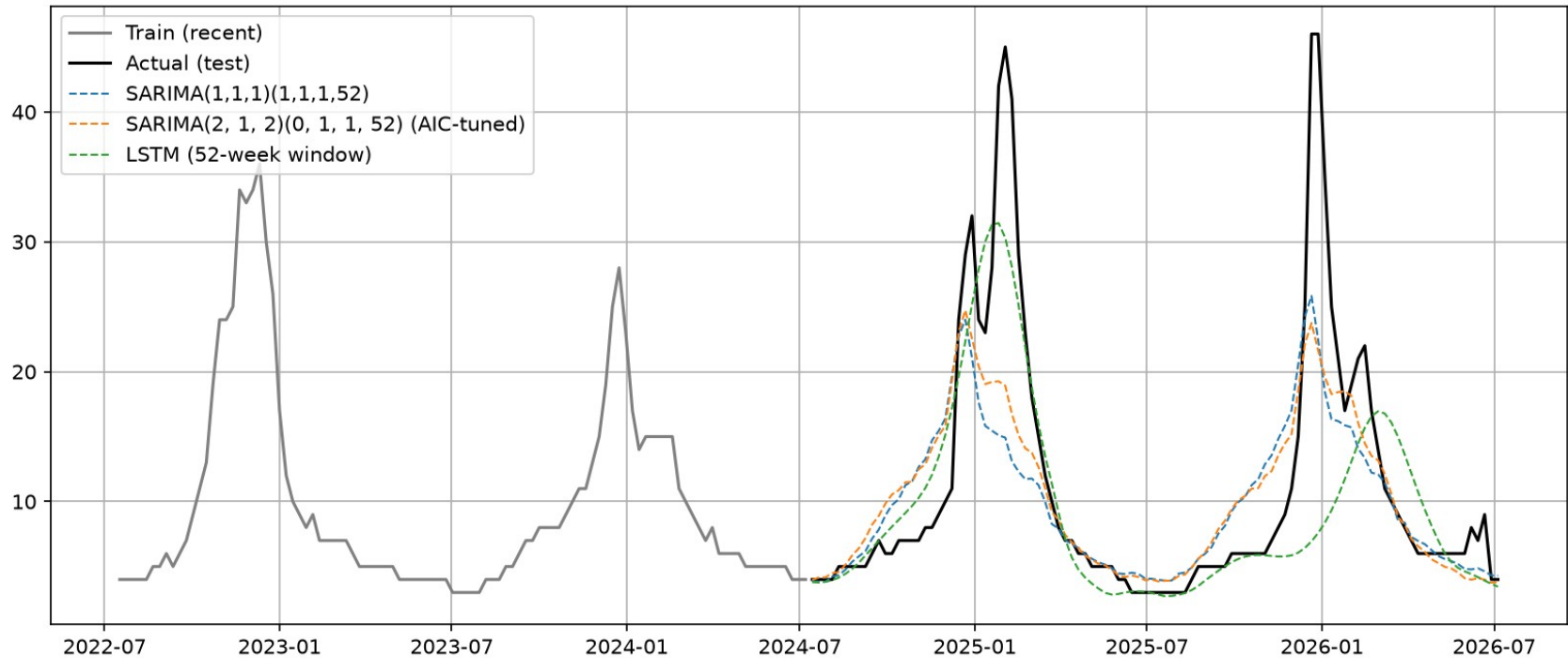
Model Selection & Training



- Classical Models – AR, MA, ARIMA, SARIMA
- Deep Learning: LSTM
- Final Model Selection : SARIMA with AIC tuned

Model Evaluation

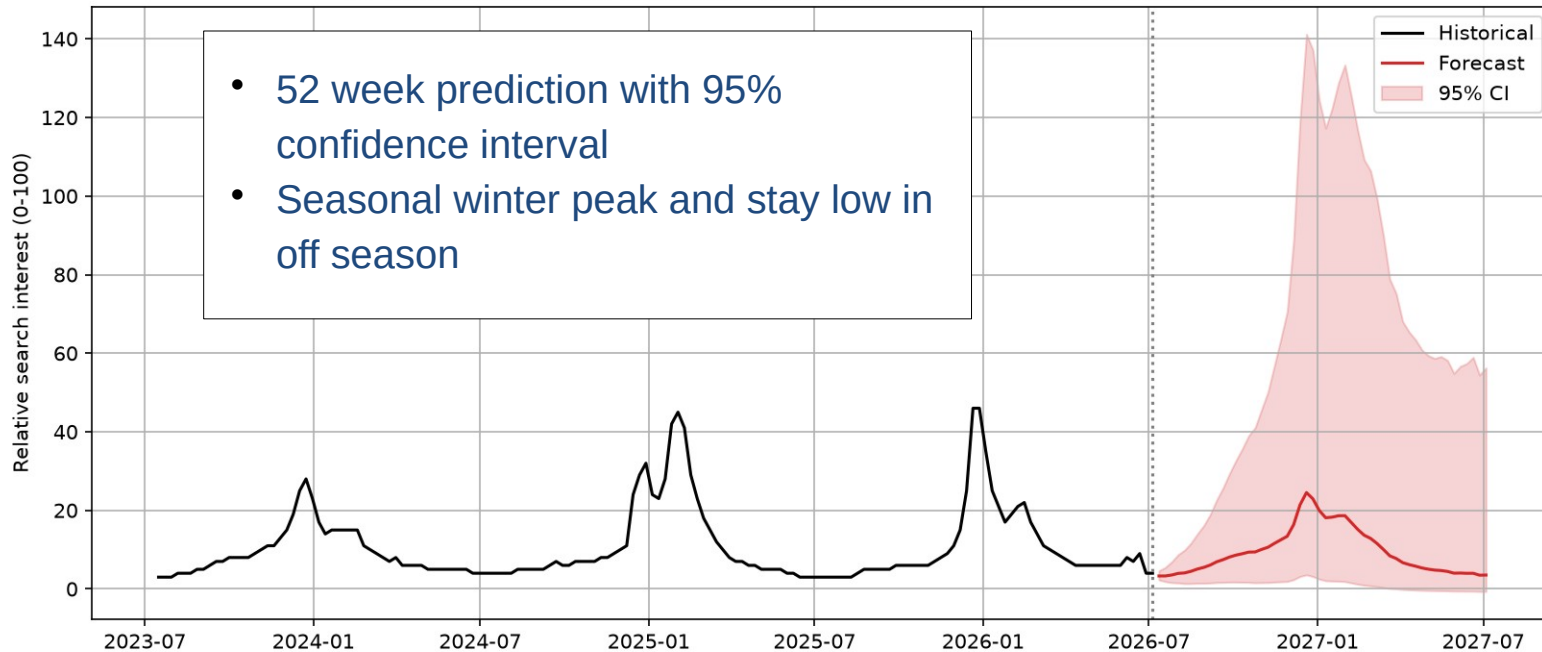
Actual vs. Forecast on Test Set



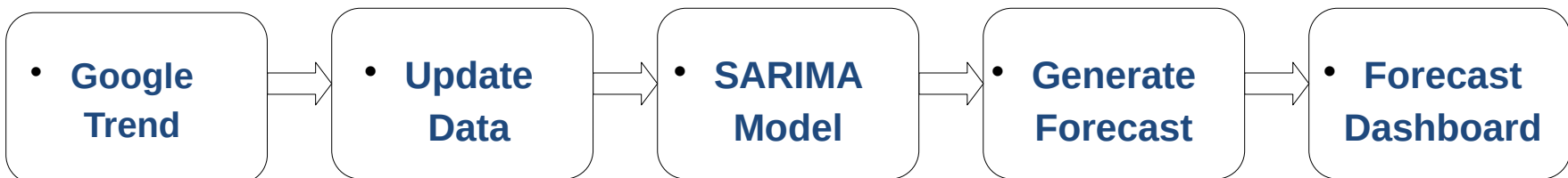
Model	MAE	RMSE
SARIMA (AIC)	3.75	6.63
SARIMA	4.48	7.12
LSTM (52 weeks)	4.45	8.14
ARIMA/ MA/ AR	7.5 ~ 7.8	12.6 ~ 12.8

Forecasting

52-Week Forecast: "flu symptoms" Search Interest (SARIMA(2, 1, 2)(0, 1, 1, 52))



Deployment



Key Findings

- Strong yearly seasonality
- Best model: AIC-tuned SARIMA — outperformed both simpler baselines and the LSTM
- Google Trends can support early flu monitoring

Limitation

- Search interest is a behavioral proxy, not a direct measurement of flu incidence
- COVID-19 caused structural changes
- Data stitching (three Google Trends windows) introduces small uncertainty
- Future work: combine with CDC data